Tutorial 3: Subject Matter Expert (SME) Report – Part 1

Project Repository: malaysiacrime
Fork Repository Link: https://github.com/ak-ayanat/malaysiacrime
Submission Date: October 2026

1. Repository Overview

- **Repository Name:** malaysiacrime (Forked to ak-ayanat/malaysiacrime)
- **Programming Language(s):** Python, HTML, CSS, JavaScript
- **Purpose of the Project:** malaysiacrime is an open-source, crowd-sourced web application designed to collect, monitor, and map crime reports across Malaysia. It was originally developed by Kegan Gan, using Django 1.3 and Python 2, at a time when public access to localized crime information in Malaysia was limited. The platform enables users to submit crime incidents (recording details such as location and timestamp), browse and search recent reports, comment on them, and subscribe to updates through RSS and sitemap feeds for public safety tracking and spatial visualization. A key feature is the crime monitor: users select an area on the map, subscribe with their e-mail address, and automatically receive notifications when new crimes are reported in that area within the last 24 hours. The codebase is organized into four Django apps: crime (crime report submission and storage), feeds (RSS feed generation), main (core pages such as the homepage, recent reports, About and Search) and monitor (area-based subscriptions and e-mail notifications). Data is stored in a relational database through the Django ORM, and schema changes are managed with South migrations. Because the project has not been maintained for many years, it no longer runs on current Python versions without significant workarounds, as shown in the build reports below: only a Python 2.7 environment, which reached end of life in 2020, can install its dependencies. This makes it a representative case study for software maintenance: it combines outdated dependencies, undocumented configuration, a limited test suite, and security practices that no longer meet current standards.

Importance of Maintaining the Project

1. **Public Safety and Community Awareness:** Maintaining a crowd-sourced crime reporting platform provides citizens and researchers with transparent access to safety data, aiding localized risk assessment and neighborhood awareness.
2. **Mitigating Critical Security Risks:** As a legacy codebase dating back over 15 years, key security features such as CSRF protection were explicitly disabled (settings_default.py). Modernizing and maintaining the code prevents vulnerabilities like Cross-Site Scripting (XSS), SQL Injection, and Cross-Site Request Forgery (CSRF).
3. **Ensuring System Longevity & Compatibility:** The application relies on deprecated Python 2 syntaxes and legacy Django standards. Updating dependencies and code structures ensures operational continuity on contemporary hosting environments and

modern web browsers.

2. Resource Identification and Maintenance Activities

Table 2: Resource Identification and Maintenance Activities

Aspect	Current Status	Evidence (link)	Issues / Key areas for improvement	Potential Maintenance Activities
Documentation	Minimal	ak-ayanat/malaysiacrime	Only contains basic contribution guidelines; lacks local setup/build steps in README.	Add a comprehensive README with installation instructions, dependencies, and deployment steps.
Design	Partially Modular	ak-ayanat/malaysiacrime	Split into crime, feeds, main, and monitor, but components share legacy Django settings.	Refactor codebase to decouple views, background monitoring tasks, and data models cleanly.
Testing	Partial	tests	View tests only (main, crime, monitor); no model/form unit tests, one app has only Django's default stub.	Add model/form unit tests and run the suite in CI.
Architecture	Monolithic	ak-ayanat/malaysiacrime	Legacy monolithic Django MVT (Model-View-Template) architecture.	Containerize the application using Docker and decouple backend APIs from frontend templates.
Source Code	Moderate	ak-ayanat/malaysiacrime	Code layout follows standard Django structure;	Format code using Black and Flake8, and add

Aspect	Current Status	Evidence (link)	Issues / Key areas for improvement	Potential Maintenance Activities
			some docstrings, inline comments, and type hints.	docstrings across functions and classes.
Code Quality	Average	ak-ayanat/malaysiacrime	Uses legacy Python 2 syntax (e.g., print statements in package scripts), breaking on Python 3.	Refactor codebase to follow modern Python 3 standards and updated framework conventions.
Dependencies	Outdated	requirements.txt	Uses obsolete library versions (Django 1.3.1, Fabric 1.2.2, Markdown 2.0.3).	Upgrade Python runtime to 3.11+, and update Django and dependent libraries to active LTS versions.
Security	Low	settings_default.py	CSRF protection is explicitly disabled (Disabled CSRF token).	Enable CSRF middleware, enforce input sanitization, and secure IP-logging routines.
Performance	Moderate	ak-ayanat/malaysiacrime	Static assets (/static) served directly without caching, CDN integration, or spatial indexing.	Implement spatial indexing on database queries and integrate CDN caching for static assets.
Version Control	Inactive	ak-ayanat/malaysiacrime	Development activity ceased over 15 years ago (229 total commits).	Resume repository maintenance, clean up obsolete branches, and

Aspect	Current Status	Evidence (link)	Issues / Key areas for improvement	Potential Maintenance Activities
				establish a version tagging system.
CI/CD Pipeline	None	ak-ayanat/malaysiacrime	Lacks build and deployment automation (no GitHub Actions or CI configuration).	Implement GitHub Actions workflows for automated linting, unit testing, and deployment checks.
User Experience (UX)	Functional	templates/	Legacy HTML templates without mobile-responsive design or interactive map features.	Redesign frontend using modern UI frameworks (e.g., Tailwind CSS) and interactive maps.
Accessibility	None	templates/	Templates lack ARIA labels, semantic roles, and keyboard navigation support.	Update HTML markup to comply with WCAG accessibility guidelines.
Internationalization	No	ak-ayanat/malaysiacrime	Hardcoded English UI strings without localization support.	Implement Django i18n framework to enable multi-language support (Bahasa Malaysia / English).
Community Engagement	None	ak-ayanat/malaysiacrime	Minimal repository interaction (2 stars, 1 fork, no active discussions).	Establish contribution guidelines, issue templates, and discussion boards for

Aspect	Current Status	Evidence (link)	Issues / Key areas for improvement	Potential Maintenance Activities
				contributors.

3. Fork and Local Development Environment

a. Fork Link

- **Fork Repository Link:** https://github.com/ak-ayanat/malaysiacrime
- **Collaborators Added:** Group members and lecturer added as repository collaborators.

b & c. Build Report

Member 1: Kanat Ayanat

i. Build Status

Status: Failed

ii. Environment Used

- **Operating System:** macOS
- **Language / Runtime Version:** Python 3.x (Anaconda base environment)
- **Build Tool / Package Manager:** pip
- **Database Required:** SQLite3 (Django default)

iii. Steps Taken to Build and Run

1. Cloned the forked repository to the local environment:
   ```
   git clone https://github.com/ak-ayanat/malaysiacrime.git
   cd malaysiacrime
   ```
2. Attempted to run the application server directly using Python:
   ```
   python manage.py runserver
   ```
3. Attempted to install project dependencies specified in requirements.txt:
   ```
   pip install -r requirements.txt
   ```

iv. Problems Encountered and Resolutions Attempted

1. **Missing Module Error (ModuleNotFoundError: No module named 'django'):**
   - *Error:* Execution of python manage.py runserver failed due to missing Django package in the environment.
   - *Attempted Resolution:* Executed pip install -r requirements.txt to install the required legacy dependencies.

2. **Build Error during Package Metadata Generation (Markdown==2.0.3 failure):**
    - *Error Message:*
      `SyntaxError: Missing parentheses in call to 'print'. Did you mean print(...)?`
      `error: metadata-generation-failed`
    - *Root Cause Analysis:* The dependency Markdown==2.0.3 specified in requirements.txt was written in Python 2 syntax (using print '...' statements without parentheses). Modern Python 3 interpreters strictly require parentheses for print() calls, causing setuptools build failures during installation.
    - *Attempted Resolution:* Attempted direct package installation, but dependency builds failed due to language syntax incompatibility between Python 2 package source code and Python 3 runtimes.

v. Screenshot of Build Error

```
Last login: Sat Jul 11 22:21:15 on ttys000
(base) admin@MacBook-Air-admin-6 ~ % git clone https://github.com/ak-ayanat/malaysiacrime.git && cd malaysiacrime
Cloning into 'malaysiacrime'...
remote: Enumerating objects: 1480, done.
remote: Total 1480 (delta 0), reused 0 (delta 0), pack-reused 1480 (from 1)
Receiving objects: 100% (1480/1480), 515.68 KiB | 11.72 MiB/s, done.
Resolving deltas: 100% (618/618), done.
(base) admin@MacBook-Air-admin-6 malaysiacrime % python manage.py runserver
Traceback (most recent call last):
  File "/Users/admin/malaysiacrime/manage.py", line 2, in <module>
    from django.core.management import execute_manager
ModuleNotFoundError: No module named 'django'
(base) admin@MacBook-Air-admin-6 malaysiacrime % pip install -r requirements.txt
Collecting Django==1.3.1 (from -r requirements.txt (line 1))
  Downloading Django-1.3.1.tar.gz (6.5 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 6.5/6.5 MB 21.3 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Collecting Fabric==1.2.2 (from -r requirements.txt (line 2))
  Downloading Fabric-1.2.2.tar.gz (155 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 155.2/155.2 kB 16.1 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Collecting Markdown==2.0.3 (from -r requirements.txt (line 3))
  Downloading Markdown-2.0.3.zip (94 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 94.6/94.6 kB 12.8 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... error
  error: subprocess-exited-with-error

  × python setup.py egg_info did not run successfully.
  │ exit code: 1
  ╰─> [7 lines of output]
      Traceback (most recent call last):
        File "<string>", line 2, in <module>
        File "<pip-setuptools-caller>", line 34, in <module>
        File "/private/var/folders/g2/jfmb8hzd6cb6t8_dkhjphw2r0000gn/T/pip-install-46cutgfx/markdown_8da0aa8fdf50426e9bacb27063d772de/setup.py", line 23
          print 'Created:', bat_path
          ^^^^^^^^^^^^^^^^^^^^^^^^^
      SyntaxError: Missing parentheses in call to 'print'. Did you mean print(...)?
      [end of output]

  note: This error originates from a subprocess, and is likely not a problem with pip.
error: metadata-generation-failed

× Encountered error while generating package metadata.
╰─> See above for output.

note: This is an issue with the package mentioned above, not pip.
hint: See above for details.
(base) admin@MacBook-Air-admin-6 malaysiacrime % python manage.py runserver
Traceback (most recent call last):
  File "/Users/admin/malaysiacrime/manage.py", line 2, in <module>
    from django.core.management import execute_manager
ModuleNotFoundError: No module named 'django'
(base) admin@MacBook-Air-admin-6 malaysiacrime %
```

Member 2: Yubin Jeong's Build Report

i. Build Status

Status: Failed

ii. Environment Used

- **Operating System:** Windows 11 (64-bit)
- **Language / Runtime Version:** Python 3.15.0
- **Build Tool / Package Manager:** pip
- **Database Required:** SQLite3 (Django default)

iii. Steps Taken to Build and Run

1. Cloned the forked repository to the local environment and navigated to the project folder:
    - `git clone [https://github.com/ak-ayanat/malaysiacrime.git](https://github.com/ak-ayanat/malaysiacrime.git)`
    - `cd malaysiacrime`
2. Attempted to run the application server directly using Python:
    - `python manage.py runserver`
3. Attempted to install project dependencies specified in requirements.txt:
    - `pip install -r requirements.txt`

iv. Problems Encountered and Resolutions Attempted

1. **Missing Module Error (`ModuleNotFoundError: No module named 'django'`):**
    - *Error:* Execution of `python manage.py runserver` failed due to missing Django package in the environment.
    - *Attempted Resolution:* Executed `pip install -r requirements.txt` to install the required legacy dependencies.
2. **Build Error during Package Metadata Generation (`Markdown==2.0.3` failure):**
    - *Error Message:*`SyntaxError: Missing parentheses in call to 'print'. Did you mean print(...)?error: metadata-generation-failed`
    - *Root Cause Analysis:* The dependency `Markdown==2.0.3` specified in `requirements.txt` was written in Python 2 syntax (using `print '...'` statements without parentheses). Running under Python 3.15.0 caused setuptools build metadata generation failures during installation.
    - *Attempted Resolution:* Package build failed at wheel requirement generation, preventing execution on modern Python 3 runtimes.

v. Screenshot of Build Error

```
PS C:\Users\j9413\OneDrive\Desktop\malaysiacrime\malaysiacrime> python --version
Python 3.15.0
PS C:\Users\j9413\OneDrive\Desktop\malaysiacrime\malaysiacrime> python manage.py runserver
Traceback (most recent call last):
  File "C:\Users\j9413\OneDrive\Desktop\malaysiacrime\malaysiacrime\manage.py", line 2, in <module>
    from django.core.management import execute_manager
ModuleNotFoundError: No module named 'django'
PS C:\Users\j9413\OneDrive\Desktop\malaysiacrime\malaysiacrime> pip install -r requirements.txt
Collecting Django==1.3.1 (from -r requirements.txt (line 1))
  Downloading Django-1.3.1.tar.gz (6.5 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 6.5/6.5 MB 771.5 kB/s  0:00:08
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Collecting Fabric==1.2.2 (from -r requirements.txt (line 2))
  Downloading Fabric-1.2.2.tar.gz (155 kB)
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Collecting Markdown==2.0.3 (from -r requirements.txt (line 3))
  Downloading Markdown-2.0.3.zip (94 kB)
  Installing build dependencies ... done
  Getting requirements to build wheel ... error
  error: subprocess-exited-with-error

  × Getting requirements to build wheel did not run successfully.
  │ exit code: 1
  ╰─> [25 lines of output]
      Traceback (most recent call last):
        File "C:\Users\j9413\AppData\Local\Python\pythoncore-3.15-64\Lib\site-packages\pip\_vendor\pyproject_hooks\_in_process\_in_process.py", line 389, in <module>
          main()
          ~~~~^^
        File "C:\Users\j9413\AppData\Local\Python\pythoncore-3.15-64\Lib\site-packages\pip\_vendor\pyproject_hooks\_in_process\_in_process.py", line 373, in main
          json_out["return_val"] = hook(**hook_input["kwargs"])
                                   ~~~~^^^^^^^^^^^^^^^^^^^^^^^^
        File "C:\Users\j9413\AppData\Local\Python\pythoncore-3.15-64\Lib\site-packages\pip\_vendor\pyproject_hooks\_in_process\_in_process.py", line 143, in get_requires_for_build_wheel
          return hook(config_settings)
        File "C:\Users\j9413\AppData\Local\Temp\pip-build-env-d4490chv\overlay\Lib\site-packages\setuptools\build_meta.py", line 333, in get_requires_for_build_wheel
          return self._get_build_requires(config_settings, requirements=[])
                 ~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
        File "C:\Users\j9413\AppData\Local\Temp\pip-build-env-d4490chv\overlay\Lib\site-packages\setuptools\build_meta.py", line 301, in _get_build_requires
          self.run_setup()
          ~~~~~~~~~~~~~~^^
        File "C:\Users\j9413\AppData\Local\Temp\pip-build-env-d4490chv\overlay\Lib\site-packages\setuptools\build_meta.py", line 520, in run_setup
          super().run_setup(setup_script=setup_script)
          ~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^
        File "C:\Users\j9413\AppData\Local\Temp\pip-build-env-d4490chv\overlay\Lib\site-packages\setuptools\build_meta.py", line 317, in run_setup
          exec(code, locals())  # noqa: S102 # exec is intentional here
          ~~~~^^^^^^^^^^^^^^^^
        File "<string>", line 23
          print 'Created:', bat_path
          ^^^^^^^^^^^^^^^^^^^^^^^^^
      SyntaxError: Missing parentheses in call to 'print'. Did you mean print(...)?
      [end of output]

  note: This error originates from a subprocess, and is likely not a problem with pip.

[notice] A new release of pip is available: 26.2.1 -> 26.2.2
[notice] To update, run: C:\Users\j9413\AppData\Local\Python\pythoncore-3.15-64\python.exe -m pip install --upgrade pip
ERROR: Failed to build 'Markdown' when getting requirements to build wheel
```

Member 3: Ezio Decadt's Build Report

i. Build Status

Status: Success.

ii. Environment Used

- **Operating System:** macOS 26.6.2 (Apple Silicon, arm64)
- **Language / Runtime Version:**
  - Attempt 1: Python 3.15.0
  - Attempt 2: Python 2.7.18 (official python:2.7 Docker image, linux/arm64)
- **Build Tool / Package Manager:** pip 26.2.1 (Attempt 1), pip 20.0.2 (Attempt 2)
- **Containerization:** Docker
- **Database Required:** SQLite3

iii. Steps Taken to Build and Run

Attempt 1: native Python 3.15 (failed)

- Cloned the forked repository and entered the project folder:
  ```
  git clone https://github.com/ak-ayanat/malaysiacrime.git
  cd malaysiacrime
  ```
- Attempted to run the server: python3 manage.py runserver
- Created and activated a virtual environment:
  ```
  python3 -m venv venv
  source venv/bin/activate.fish
  ```
- Attempted to install dependencies: `pip install -r requirements.txt`

Attempt 2: Python 2.7 in Docker (success)

- Started a Python 2.7 container, mounting the project into a folder named malaysiacrime:
  ```
  docker run -it --rm -v "$PWD":/src/malaysiacrime -w
  /src/malaysiacrime -p 8000:8000 python:2.7 bash
  ```
- Installed all dependencies: `pip install -r requirements.txt`. All seven packages built successfully: Django 1.3.1, Fabric 1.2.2, Markdown 2.0.3, South 0.7.3, ipython 0.11, paramiko 1.7.7.1 and pycrypto 2.3.
- Created a local settings.py that imports settings_default.py and overrides the database configuration:
  ```
  from settings_default import *
  DEBUG = True
  TEMPLATE_DEBUG = True
  SECRET_KEY = 'local-dev-key-not-for-production'
  DATABASE_ENGINE = 'sqlite3'
  DATABASE_NAME = os.path.join(PROJECT_DIR, 'dev.db')
  DATABASES = {'default': {'ENGINE': 'django.db.backends.sqlite3',
  'NAME': os.path.join(PROJECT_DIR, 'dev.db')}}
  MEDIA_URL = '/static/'
  ```
- Created the database tables and applied the South migrations:
  ```
  python manage.py syncdb --noinput
  python manage.py migrate
  ```

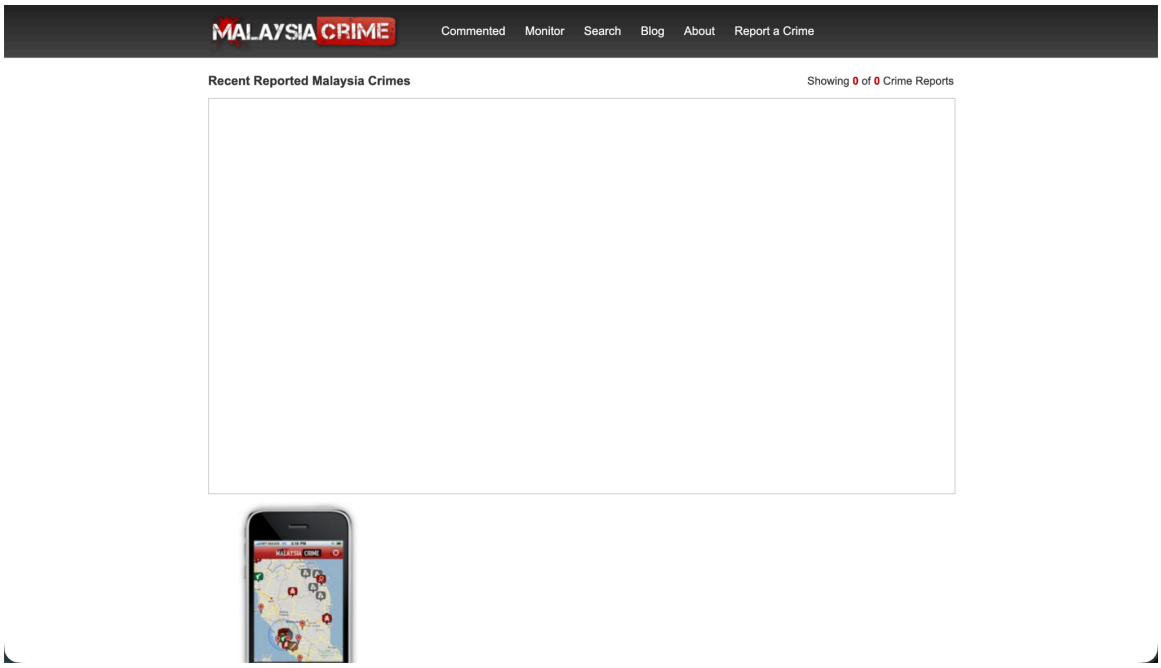
- Started the development server, binding to all interfaces so it is reachable from the host: `python manage.py runserver 0.0.0.0:8000`
- Opened `http://localhost:8000` in the browser on macOS.

iv. Problems Encountered and Resolutions

4. **Missing Django module**
   Error: ModuleNotFoundError: No module named 'django'
   Note: The traceback points to from django.core.management import execute_manager. This function was removed in Django 1.6, so manage.py is incompatible with any modern Django release.
   Resolution: Installed the dependencies (see below).
5. **Markdown 2.0.3 fails to build on Python 3**
   Error: `SyntaxError: Missing parentheses in call to 'print'. Did you mean print(...)?` followed by `ERROR: Failed to build 'Markdown' when getting requirements to build wheel`
   Root Cause: The `setup.py` of Markdown 2.0.3 uses Python 2 syntax (`print 'Created:', bat_path`), which is invalid in Python 3.15. This confirms the failure reported by the other members.
   Resolution: Switched to Python 2.7, the runtime the project was written for. No native Python 2.7 build is available for Apple Silicon through Homebrew, so the official `python:2.7` Docker image was used (Attempt 2).
6. **No settings file and no database configured**
   Problem: The repository only ships settings_default.py, with an empty DATABASE_ENGINE and DATABASE_NAME. No settings.py exists, and the README gives no setup instructions.
   Resolution: Created a local settings.py that imports the defaults and configures SQLite (step 7).
7. **Project package name mismatch**
   Error: ImproperlyConfigured: Error importing middleware malaysiacrime: "No module named malaysiacrime"
   Root Cause: The project uses the pre-Django 1.4 layout, where imports are prefixed with the project folder name (malaysiacrime.*). The code was first mounted at /app, so Django loaded app.settings instead.
   Resolution: Remounted the project at /src/malaysiacrime and reinstalled the dependencies in the new container.
8. **Static files (CSS and images) not loading**
   Problem: The homepage rendered as unstyled HTML, and the logo and other images appeared broken.
   Root Cause: `MEDIA_URL` is empty (`''`) in `settings_default.py`. The templates therefore generated relative links such as `stylesheets/base.css`, while Django's development server only serves static files under `/static/`.
   Resolution: Added `MEDIA_URL = '/static/'` at the end of the local `settings.py`. This file is ignored by git, so the repository itself is unchanged. The page now requests `/static/stylesheets/base.css` and `/static/images/logo.gif`, and `curl` on the stylesheet returns HTTP 200. The development server reloaded the configuration automatically, so a page refresh was enough.
   Note: The homepage shows "Showing 0 of 0 Crime Reports" because the new `dev.db`

database contains no records. This is expected and does not indicate an error.

v. Screenshots



Member 2: Student's Build Report

i. Build Status: Success

ii. Environment Used:

3. **OS:** macOS
4. **Language / Runtime Version:** Python 2.7.18 (managed via pyenv)
5. **Build Tool / Package Manager:** pip (<21)
6. **Database Required:** SQLite3 (Django default)

iii. Steps Taken to Build and Run:

2. Configured legacy Python environment using pyenv:
   ```
   brew install pyenv
   pyenv install 2.7.18
   pyenv local 2.7.18
   ```
3. Set up application configuration file:
   `cp settings_default.py settings.py`Configured SQLite as the database engine in settings.py:
   ```
   DEBUG = True
   TEMPLATE_DEBUG = True
   DATABASE_ENGINE = 'sqlite3'
   DATABASE_NAME = 'dev.db'
   ```
4. Installed legacy dependencies using downgraded pip:
   ```
   python -m pip install --upgrade "pip<21"
   python -m pip install -r requirements.txt
   ```
5. Initialized database and launched application server:
   ```
   python manage.py syncdb
   python manage.py migrate
   python manage.py runserver
   ```

iv. Problems Encountered and Resolutions Attempted:

2. **Python 3 Syntax Incompatibility with Legacy Packages (Markdown==2.0.3):**
   - *Error:* Initial attempts on modern Python environments fail with SyntaxError: Missing parentheses in call to 'print'.
   - *Resolution:* Bypassed Python 3 installation issues by configuring a dedicated legacy **Python 2.7.18** environment using pyenv.
3. **Missing Application Settings File (settings.py):**
   - *Error:* Running python manage.py directly throws an error due to the absence of settings.py.
   - *Resolution:* Manually copied settings_default.py to settings.py and declared the SQLite database engine configuration.

4. **Incompatible Modern pip Version:**
    - *Error:* Newer versions of pip fail to resolve dependencies built for legacy Python 2 packages.
    - *Resolution:* Downgraded pip to a pre-v21 release (pip<21), enabling legacy wheel and setup resolution.

v. Screenshot of Running Application:



- Commented
- Monitor
- Search
- Blog
- About
- Report a Crime

Recent Reported Malaysia Crimes

Showing **0** of **0** Crime Reports



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